

GULFPORT-BILOXI, MS, USA

WMO: 747570

Lat: 30.412N

Lon: 89.081W

Elev: 42

StdP: 14.67

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 93874

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.4	32.8	14.3	11.4	37.6	18.9	14.3	40.3	22.7	52.3	20.5	53.0	4.8	350	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	93.9	77.9	92.1	77.7	90.6	77.5	80.9	88.8	80.2	87.9	79.4	87.1	7.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	151.4	84.2	78.3	147.1	83.9	77.3	142.0	83.6	44.4	88.9	43.5	87.2	42.8	86.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	18.0	16.1	23.2	98.2	3.3	2.1	20.8	99.7	18.9	100.9	17.0	102.1	14.6	103.6		
			20.7	82.9	3.3	1.6	18.3	84.1	16.3	85.0	14.5	86.0	12.1	87.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.5	51.5	55.3	61.0	68.0	75.2	81.1	82.6	82.5	79.1	70.1	60.0	54.3
	DBStd	13.06	9.61	8.85	7.78	6.21	5.13	3.13	2.63	2.68	4.24	7.33	8.11	9.12
	HDD50	223	99	43	10	0	0	0	0	0	0	0	13	58
	HDD65	1498	425	286	168	41	3	0	0	0	0	37	189	349
	CDD50	6968	147	193	352	541	781	932	1012	1008	874	624	314	190
	CDD65	2769	8	16	45	132	319	482	547	543	425	196	40	16
	CDH74	26939	4	20	127	704	2916	5267	6103	5959	4103	1566	156	14
	CDH80	10089	0	0	8	64	761	2149	2664	2546	1527	361	9	0
Wind	WSAvg	6.5	7.3	7.5	7.8	8.0	7.1	5.6	4.7	4.9	6.0	6.3	6.3	7.1
Precipitation	PrecAvg	63.7	5.4	4.5	6.0	5.4	5.0	6.1	6.9	6.1	6.2	3.0	4.0	5.0
	PrecMax	86.2	20.8	12.6	11.7	13.0	14.2	16.7	12.8	14.5	26.0	8.7	12.6	16.7
	PrecMin	41.3	0.0	0.4	0.1	0.1	0.4	0.7	1.9	1.5	0.8	0.0	0.0	1.3
	PrecStd	12.0	4.3	2.8	3.2	3.7	3.5	3.7	3.0	2.8	5.1	2.4	3.0	3.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	76.9	81.0	84.0	91.5	95.4	96.9	96.7	94.2	89.3	81.3	75.7
		MCWB	66.1	68.2	67.2	69.9	74.1	77.3	78.9	78.0	76.5	75.4	70.9	69.4
	2%	DB	71.0	73.2	77.3	81.8	88.1	92.6	93.5	93.3	91.1	86.2	78.4	73.0
		MCWB	65.3	67.7	67.6	70.1	73.7	77.1	78.2	78.8	76.6	74.3	69.5	68.2
	5%	DB	68.5	71.0	74.7	80.1	85.9	90.6	91.6	91.3	89.5	83.9	75.3	70.8
		MCWB	64.6	66.4	66.7	69.7	73.7	76.8	78.1	78.3	76.3	73.0	67.7	67.0
	10%	DB	66.2	68.4	72.5	78.4	84.0	88.8	90.2	90.1	87.7	81.7	73.1	68.5
		MCWB	63.1	64.2	66.1	69.4	73.3	76.4	77.8	78.2	76.1	72.4	66.7	65.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.2	72.1	72.1	76.0	78.5	81.1	82.0	82.4	80.7	79.9	75.1	72.0
		MCDB	71.2	74.6	76.0	80.0	84.5	88.8	90.5	90.6	86.7	84.3	78.3	74.4
	2%	WB	67.3	69.4	70.6	74.4	77.4	80.1	80.5	81.1	79.6	78.0	72.4	70.0
		MCDB	69.3	71.7	74.2	78.7	83.5	87.8	89.0	89.2	86.1	83.0	75.7	72.0
	5%	WB	65.9	67.3	69.3	73.2	76.3	79.1	79.8	80.2	78.6	76.4	70.4	68.0
		MCDB	67.6	69.7	72.9	77.5	82.7	86.8	88.4	88.0	85.3	81.7	73.8	69.9
	10%	WB	63.9	65.4	67.8	71.9	75.3	78.2	79.0	79.4	77.9	74.9	68.6	65.9
		MCDB	65.7	67.4	71.5	76.1	81.9	85.7	87.3	86.9	84.6	79.9	72.3	68.0
Mean Daily Temperature Range	5% DB	MDBR	18.3	17.5	17.8	16.9	16.1	14.8	15.0	15.1	15.9	19.1	20.3	17.8
		MCDBR	15.9	14.9	16.8	16.9	17.0	16.9	17.2	16.8	17.1	18.2	18.7	15.3
		MCWBR	11.4	10.2	9.6	9.0	7.2	6.3	6.3	6.4	6.7	8.6	11.7	11.4
	5% WB	MCDBR	12.8	13.2	13.1	12.3	12.8	13.9	15.1	15.0	13.9	13.1	14.6	13.8
		MCWBR	10.4	10.7	9.4	8.4	6.7	6.2	6.6	6.6	6.5	7.9	11.2	11.8
Clear-Sky Solar Irradiance	taub	0.334	0.350	0.373	0.405	0.438	0.466	0.516	0.505	0.454	0.385	0.354	0.341	
	taud	2.539	2.500	2.428	2.342	2.283	2.249	2.136	2.165	2.269	2.452	2.483	2.531	
	Ebn at Noon	285	290	289	282	272	263	249	251	260	274	275	277	
	Edh at Noon	27	31	36	40	43	44	49	47	41	32	29	26	
All-Sky Solar Radiation	RadAvg	957	1131	1522	1859	2048	1985	1872	1774	1654	1441	1101	850	
	RadStd	80	158	126	152	113	151	134	135	113	153	97	74	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.28	N/A	N/A	+1.39	N/A	N/A	N/A	N/A	+410	+259
(47)	Regional (28 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+167

Nomenclature: See separate page